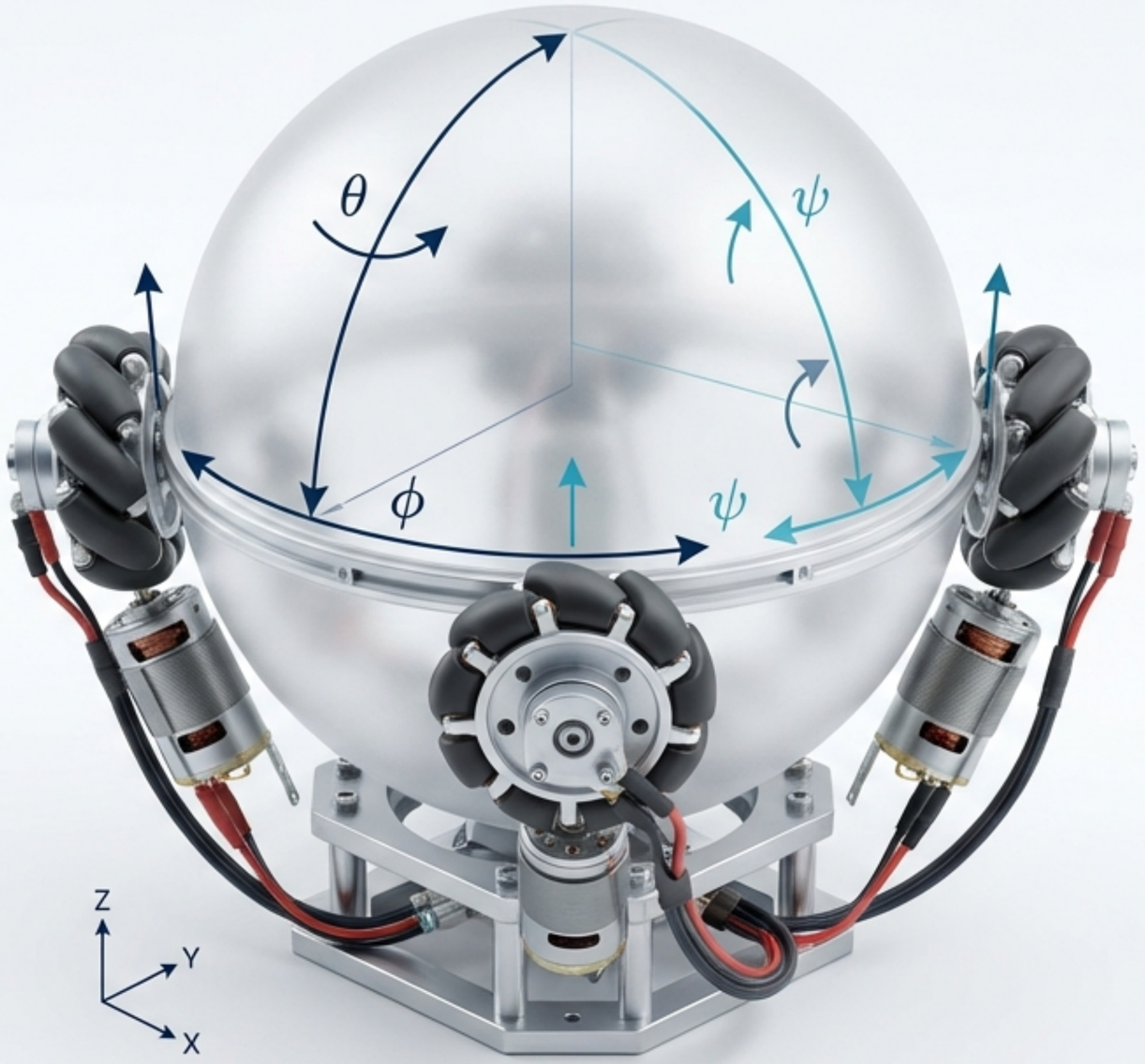
Real-Time Control Architecture for a Spherical Parallel Manipulator

Simulink & Quarc-Based Implementation of Closed-Loop Inverse Kinematics

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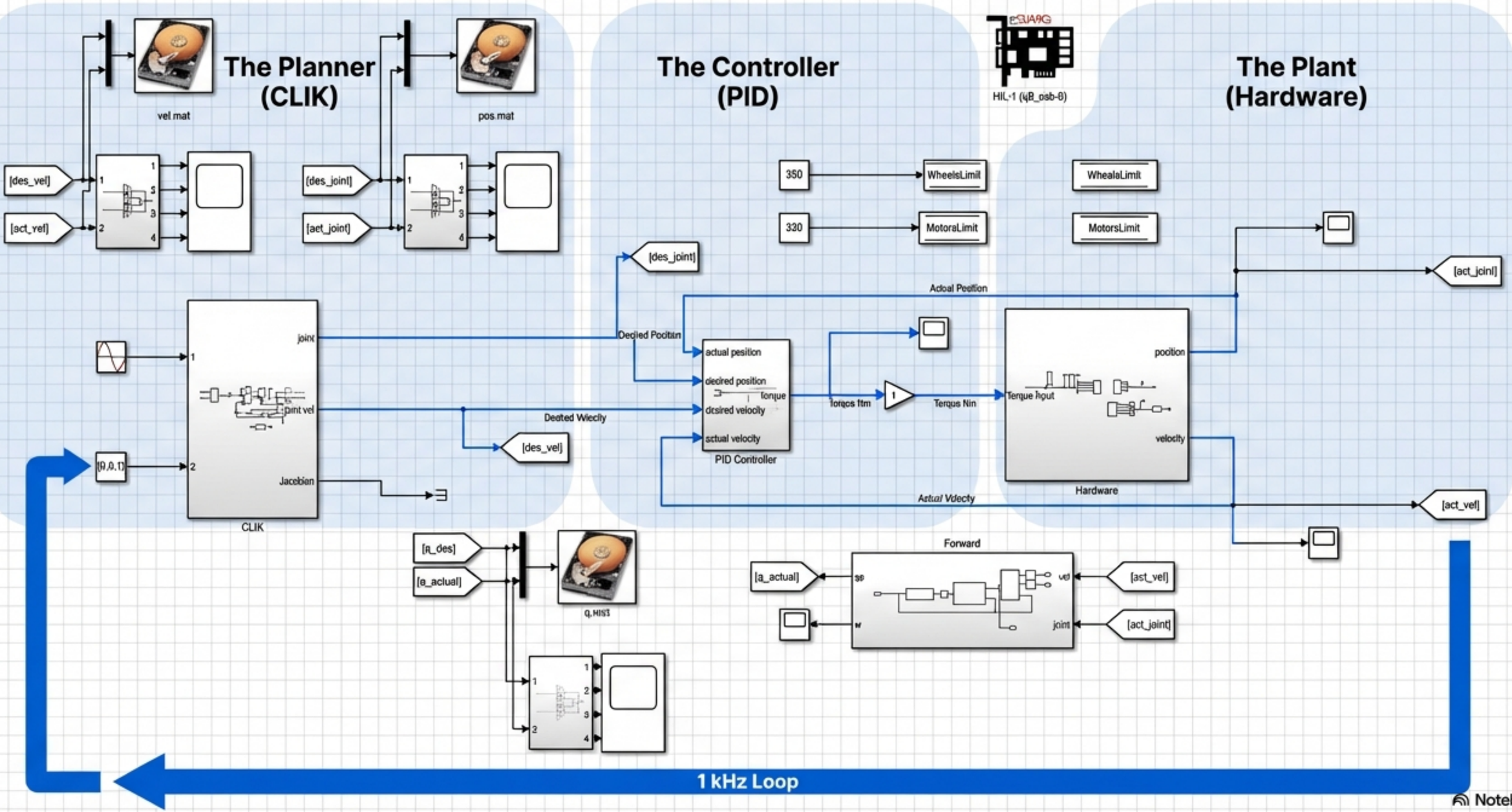


Technical Specifications

System:	Holonomic Spherical Parallel Manipulator
Actuation:	4x DC Motors (Redundant Configuration)
Control Freq:	1 kHz (Hard Real-Time)
Interface:	Quanser Q8-USB / Quarc

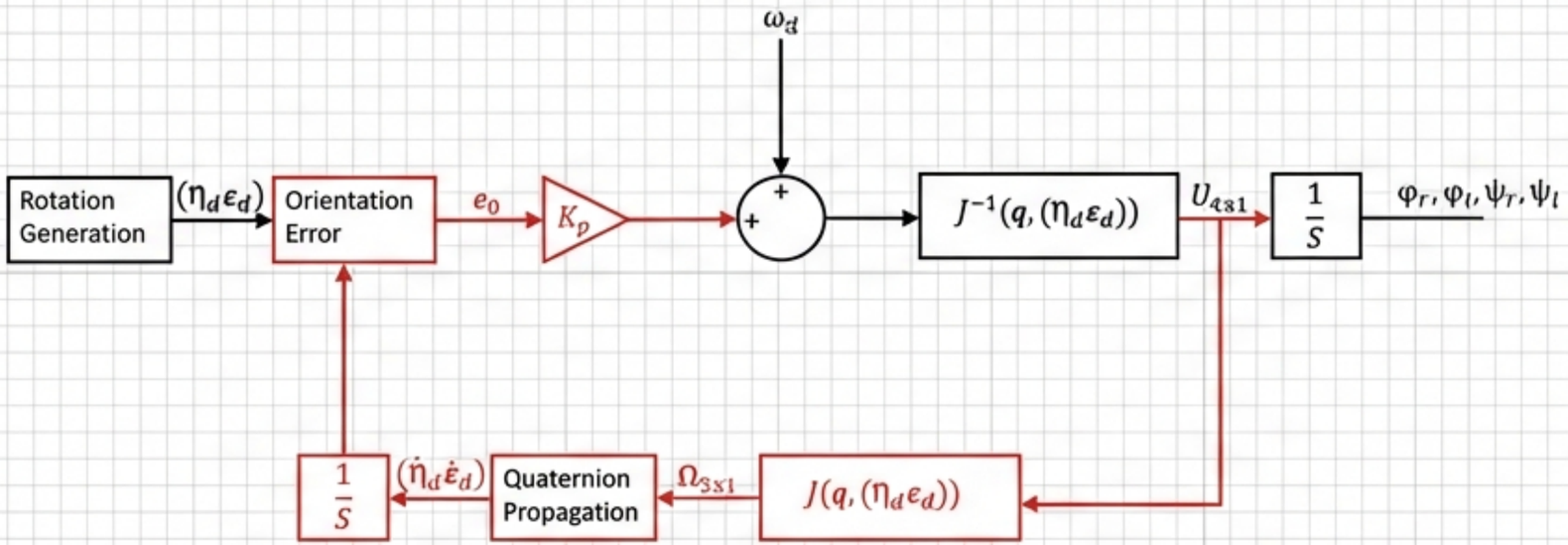
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The Control Signal Pipeline: Input-to-Actuation



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Closed-Loop Inverse Kinematics (CLIK)



The Problem (Open Loop):

$$\dot{q} = J^{-1}\omega$$

Vulnerable to numerical drift over time.

The Solution (CLIK):

$$\dot{q} = J^{-1}(v_d + K_p e_o)$$

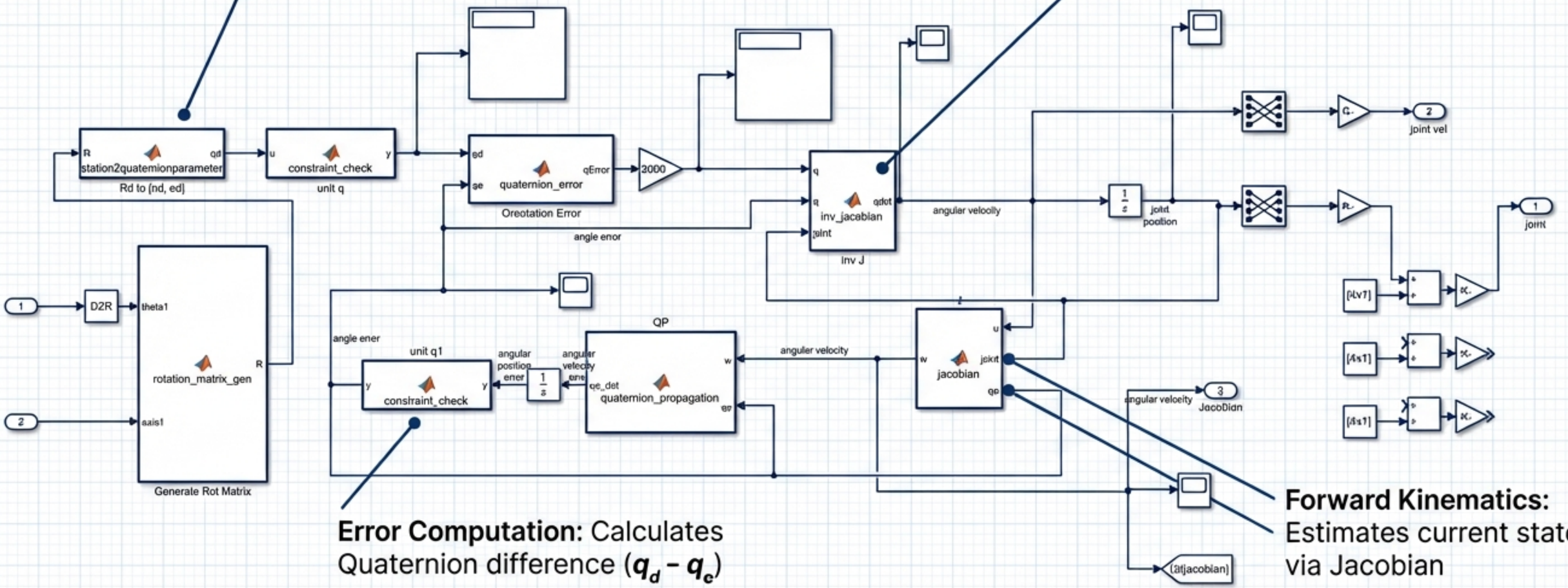
Corrects orientation error (e_o) before integration.

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Inside the CLIK Subsystem

Reference Gen: Converts trajectory to Rotation Matrix

Projection: Maps error to joint velocity space



Error Computation: Calculates Quaternion difference ($q_d - q_e$)

Forward Kinematics: Estimates current state via Jacobian

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Mathematical Deep Dive: The Forward Jacobian

The Code

```
% Optimization via Z-variables
z(10) = cos(phi_r);
z(11) = sin(phi_r);
z(18) = -z(10)^2 - z(11)^2;
z(6)  = 2*q2*q3 - 2*q1*q4;
...
w(1) = ...
w(1) = ... % Populating Matrix
```

The Matrix

$$J = \begin{bmatrix} \frac{\partial \omega_x}{\partial \dot{q}_1} & \cdots & \frac{\partial \omega_x}{\partial \dot{q}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \omega_z}{\partial \dot{q}_1} & \cdots & \frac{\partial \omega_z}{\partial \dot{q}_n} \end{bmatrix}$$

Real-time Optimization: The Jacobian calculation is flattened into a vector of intermediate variables (***z***) to minimize trigonometric calls during the 1ms interrupt cycle.

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Solving for Redundancy: The Inverse Jacobian

Distributing 3-DOF Task Space commands to 4 Actuators

```
% Pseudo-Inverse Solution for 4 Motors (u4, u5, u6, u7)
u6 = 57.58 * z(104)*u1 + 57.58  * z(105)*u2 + ...
u7 = -57.58 * z(110)*u1 - 57.58 * z(111)*u2 - ...

= [u4; u5; u6; u7];
qdot = [u4; u5; u6; u7];
```

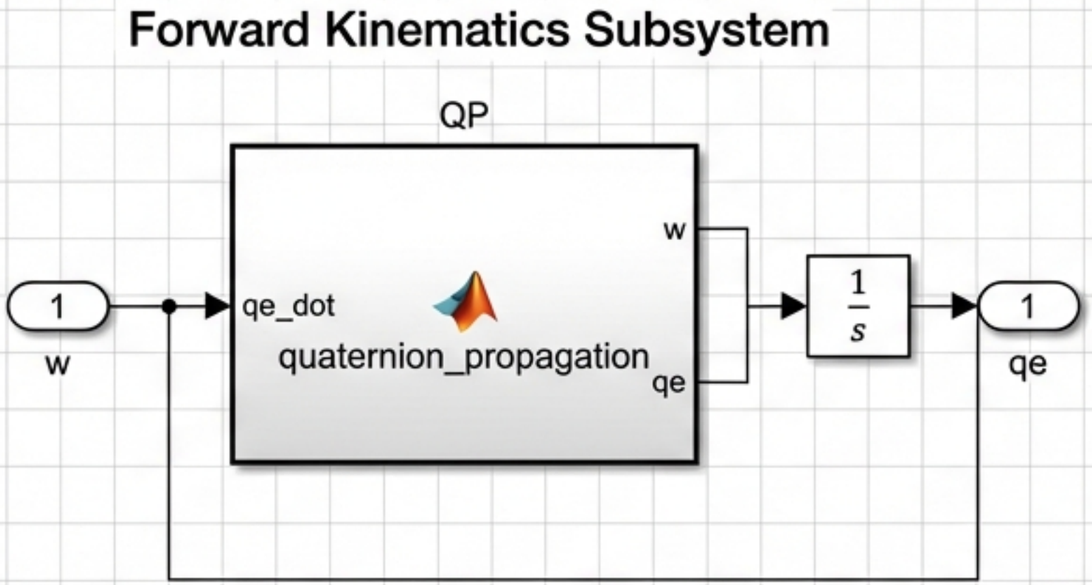
Scaling Factors

Geometry
Geometry Coefficients

The system is over-actuated (4 motors, 3 DOFs). The pseudo-inverse solution distributes torque to minimize internal fighting between motors.

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Quaternion Propagation Dynamics



$$\dot{\eta} = \frac{1}{2}(-\omega_1\epsilon_1 - \omega_2\epsilon_2 - \omega_3\epsilon_3)$$

$$\dot{\epsilon} = \frac{1}{2}(\omega \times \epsilon + \eta\omega)$$

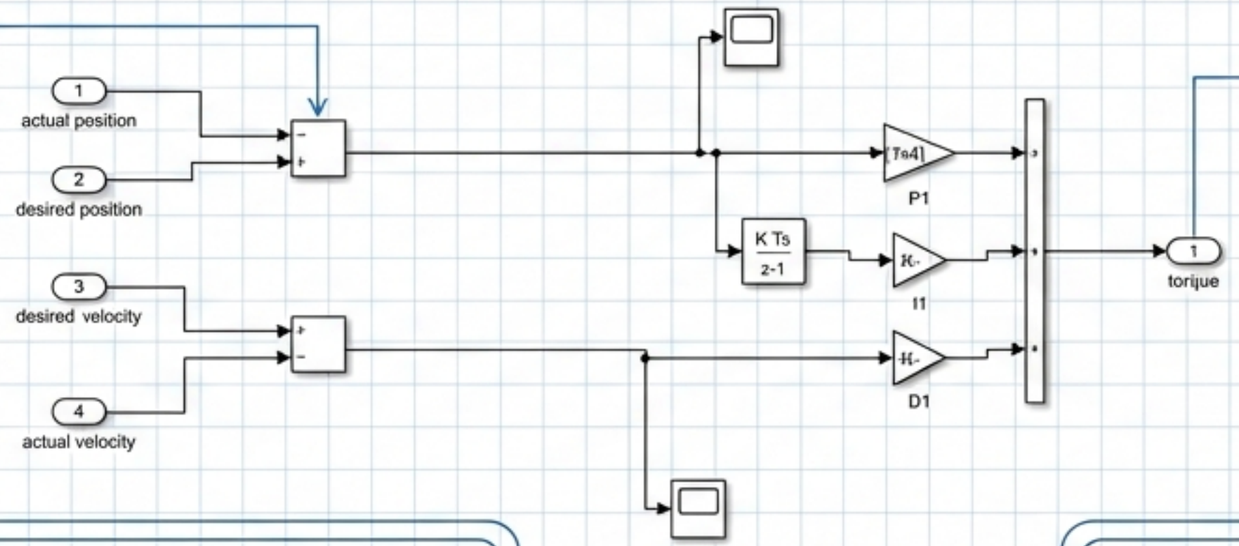
Integration in Quaternion Space avoids Gimbal Lock singularities inherent in Euler Angle integration.

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From Kinematics to Dynamics: PID Control

Controller Input

Position Error + Velocity Error



Controller Output

Torque (Nm)

Tuning Strategy

Large Motors (BML/BMR): PD Control Only. Integral gain removed to suppress oscillation.

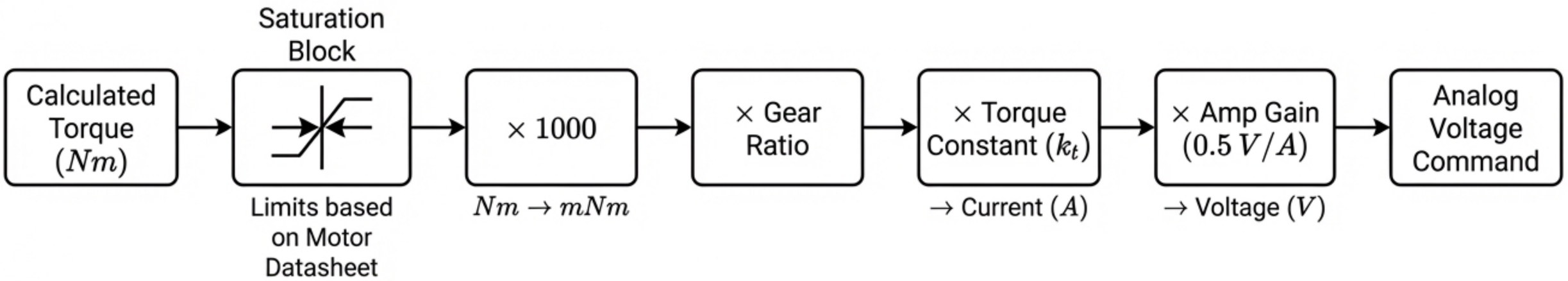
Small Motors (SML/SMR): PID Control. Full integral term required for tracking.

Asymmetry

SMR required K_p 4x larger than SML due to hardware variances.

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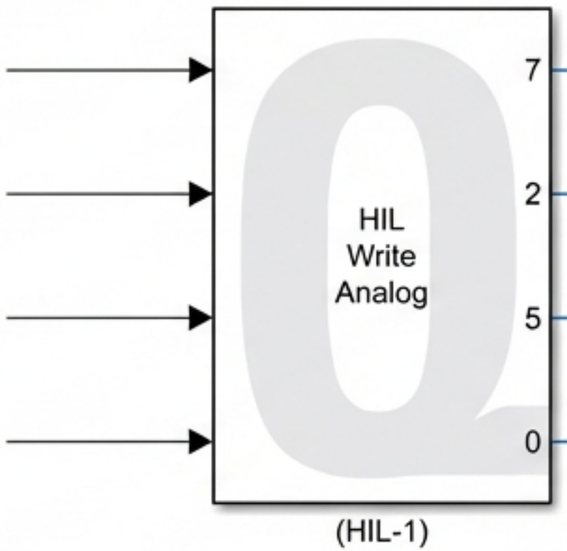
Torque Signal Conditioning



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The Quarc Hardware Interface (HIL Write)

Bridging Simulink to the Physical World



Analog Output Channels to Motor Drivers (Channels 7, 2, 5, 0)

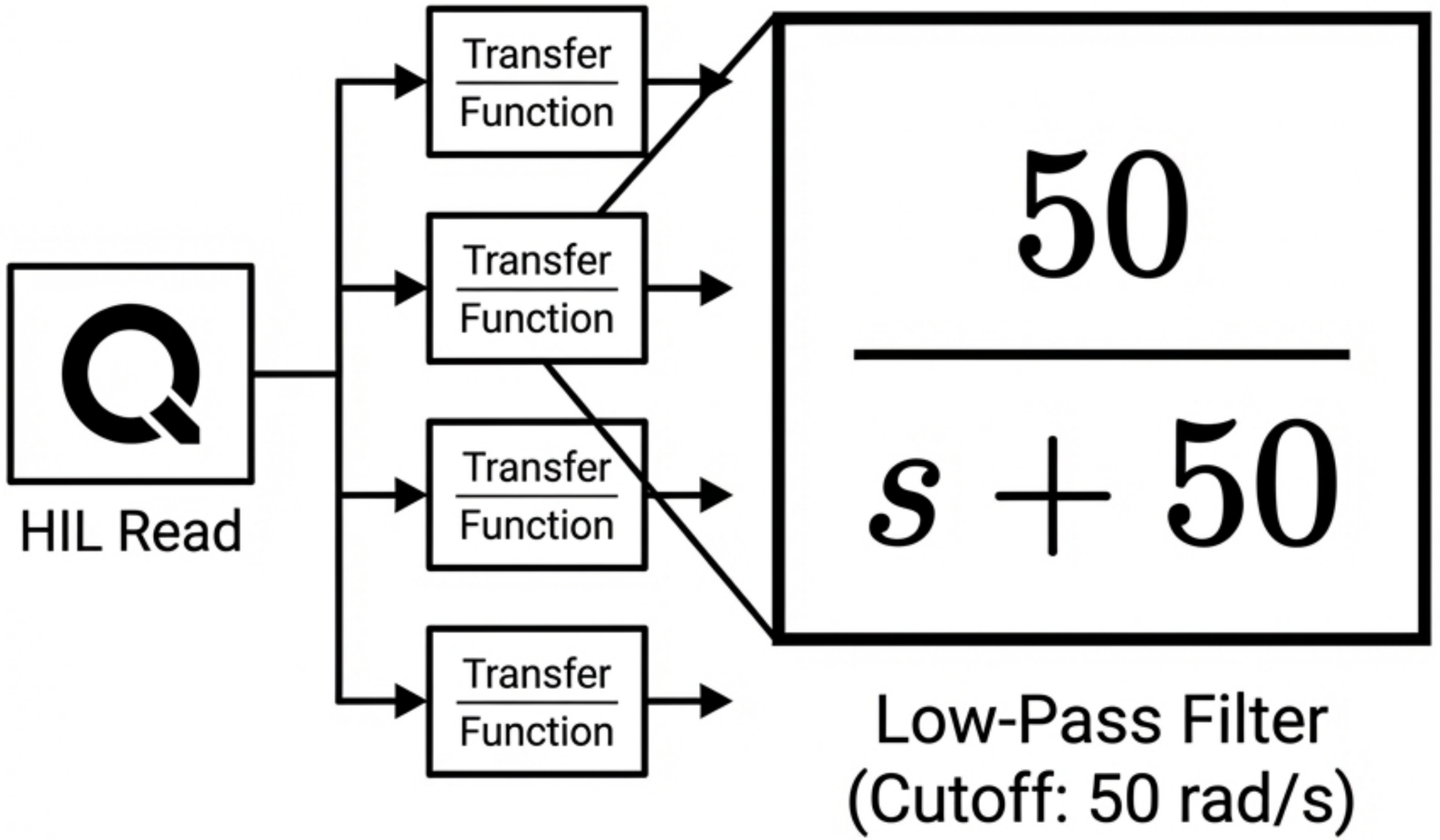
Hardware: Quanser Q8-USB Data Acquisition Board

Sample Rate: 1 kHz Update Frequency

The HIL Write block intercepts the conditioned voltage signal and triggers the Digital-to-Analog (D/A) converters on the Q8-USB board.

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Feedback Acquisition & Filtering

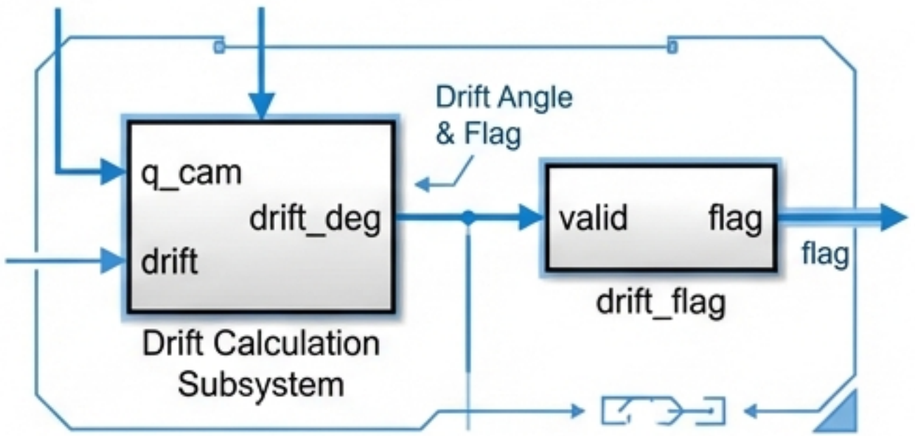


1. **HIL Read:** Acquire raw encoder counts.
2. **Gain:** Convert counts to radians.
3. **Filter:** Suppress quantization noise for clean derivative (D) term.

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Drift Detection Logic

$$Drift\ Error = \| q_{encoder} - q_{camera} \|$$



- Drift < 6° (Normal Operation)
- Drift 6° – 10° (Warning)
- Drift ≥ 10° (System Stop / ALARM)

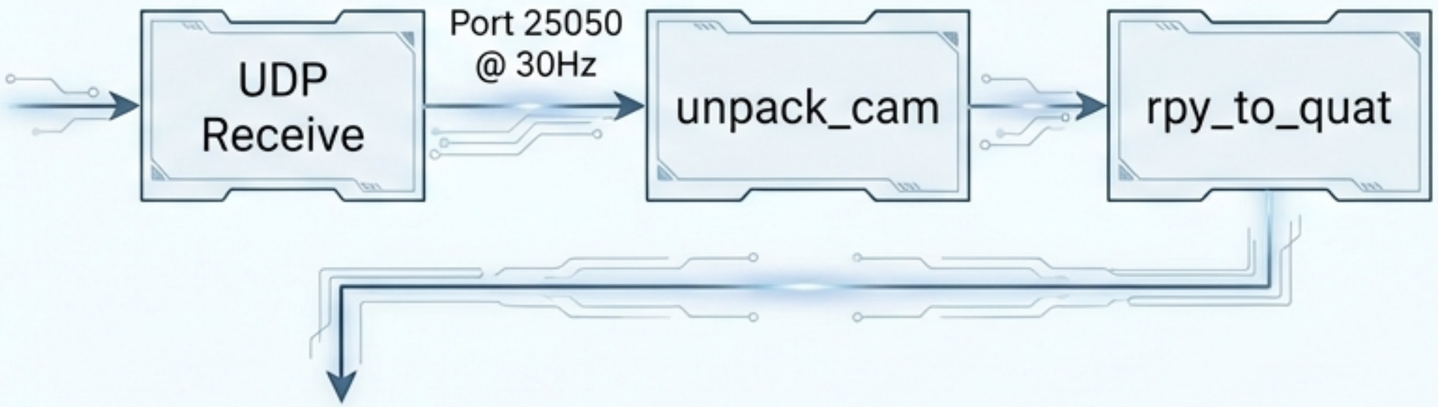
Since encoders cannot detect ball slip, the system compares the internal Kinematic model against an external Optical Sensor truth.

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External Sensing & Communication

UDP Packet Structure

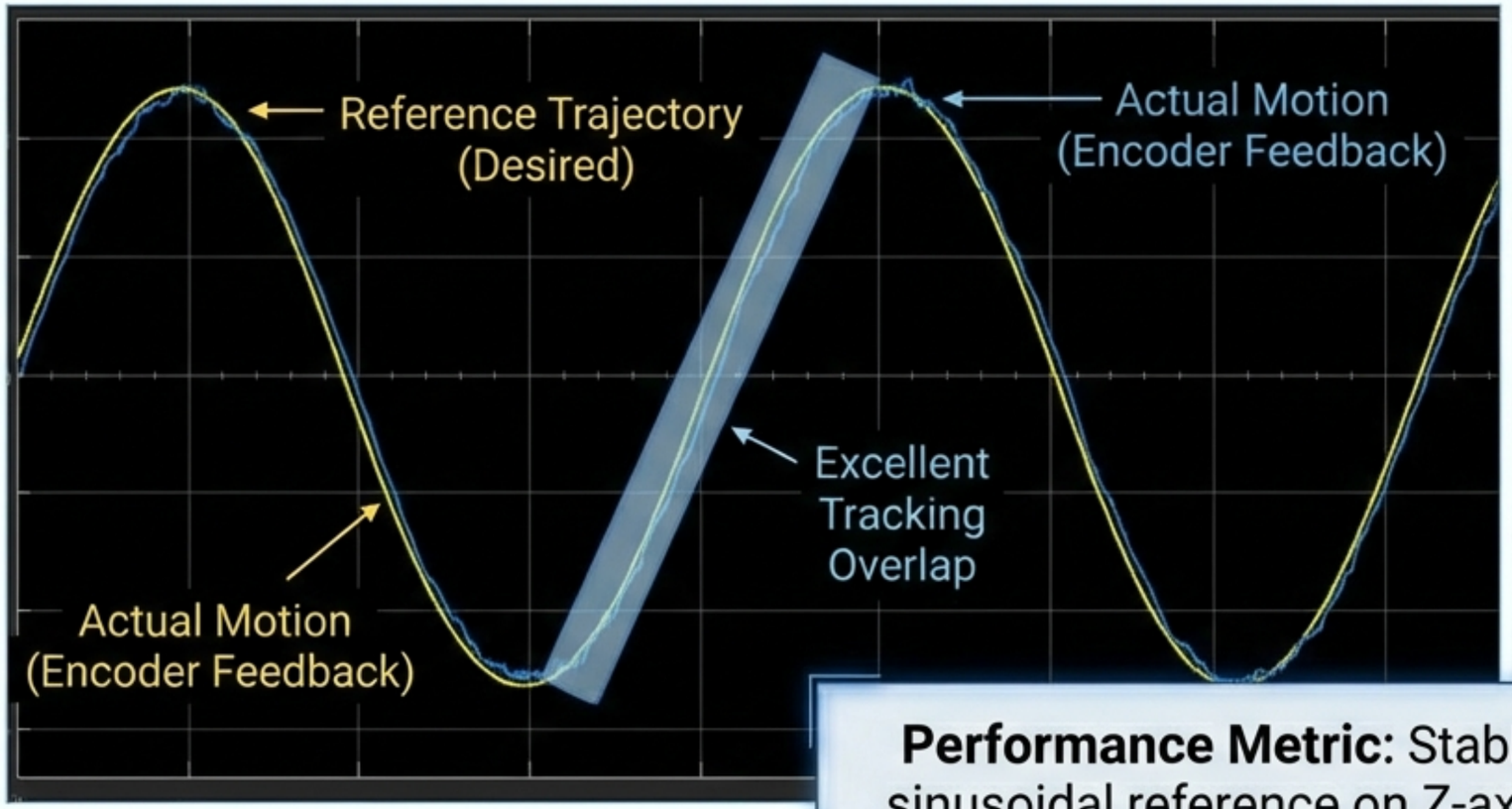
```
{
  "roll": 12.34,
  "pitch": -5.67,
  "yaw": 89.01,
  "fps": 30
}
```



Optical mouse sensor data is ingested asynchronously via UDP, unpacked, and converted to Quaternions for alignment with the control frame.

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System Validation & Performance



Performance Metric: Stable tracking of sinusoidal reference on Z-axis. PID tuning on small motors eliminated high-frequency oscillation seen in initial tests.

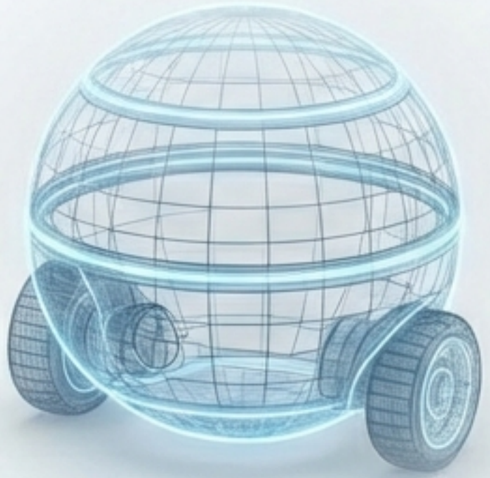
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Conclusion & Future Outlook

Summary

- ✓ **Real-Time Control**: Successfully implemented 1kHz CLIK loop on Quarc hardware.
- ✓ **Math**: Validated Quaternion-based formulation for singularity avoidance.
- ✓ **Sensing**: Established UDP pipeline for optical drift detection.

Future Work



- **Sensor Fusion**: Close the loop: Feed camera data directly into CLIK ($v_d + K_p e_{camera}$).
- **Null Space Control**: Exploit the 4th motor to optimize manipulability.

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